Unit 6: Multivariable Calculus Foundations

Prepared by Staples Education

Practice Set: Problems and Complete Solutions

Part II — Review Guide: Practice Problems

Work the following ten problems in order. They climb from first and mixed partial derivatives, through the gradient and directional derivatives, into the multivariable chain rule, and close with a full optimization sweep — finding and classifying critical points with the second-derivative test. Solutions follow in Part III.

1. For $f(x,y) = x^2y + y^3$, compute both first partials f_x and f_y.
2. For $f(x,y) = x^2y^3$, compute the mixed partials f_{xy} and f_{yx} and confirm Clairaut's theorem.
3. For $f(x,y) = x^2 + xy + y^2$, find the gradient ∇f and evaluate it at the point $(1,2)$.
4. Using the gradient from Problem 3, compute the directional derivative of f at $(1,2)$ in the direction of $\mathbf{v} = (3,4)$. Remember to normalize.
5. At the same point $(1,2)$, state the direction of steepest ascent of $f = x^2 + xy + y^2$ and the maximum rate of increase there.
6. For $z = x^2 + y^2$ with $x = \cos t$ and $y = \sin t$, use the chain rule to find $\dfrac{dz}{dt}$, and explain the result geometrically.
7. For $z = x^2y$ with $x = s + t$ and $y = st$, use the chain-rule tree to find $\dfrac{\partial z}{\partial s}$.
8. Find and classify the single critical point of $f(x,y) = x^2 + y^2 - 4x - 6y + 13$, and give the extreme value.
9. Find and classify the critical point of $f(x,y) = x^2 - y^2$ using the second-derivative test.
10. **(Capstone)** Find *all* critical points of $f(x,y) = x^3 - 3x + y^2$ and classify each with the second-derivative test, reporting any extreme values.

Part III — Complete Solutions

Solution 1. Hold y constant for f_x, then x constant for f_y. The pure-y term y^3 vanishes under ∂_x:

$$f_x = 2xy, \qquad f_y = x^2 + 3y^2.$$

Solution 2. Differentiate in x first, then y — and in the reverse order:

$$f_x = 2xy^3 \;\Rightarrow\; f_{xy} = 6xy^2, \qquad f_y = 3x^2y^2 \;\Rightarrow\; f_{yx} = 6xy^2.$$

Both equal $6xy^2$, so $f_{xy} = f_{yx}$ — Clairaut's theorem holds because the second partials are continuous everywhere.

Solution 3. Differentiate componentwise:

$$\nabla f = (2x + y,\; x + 2y), \qquad \nabla f(1, 2) = (2 + 2,\; 1 + 4) = (4, 5).$$

Solution 4. The raw direction $\mathbf{v} = (3, 4)$ has $|\mathbf{v}| = \sqrt{9 + 16} = 5$, so the unit direction is $\mathbf{u} = (\frac{3}{5}, \frac{4}{5})$. Then

$$D_{\mathbf{u}}f = \nabla f(1, 2) \cdot \mathbf{u} = (4, 5) \cdot \left(\tfrac{3}{5}, \tfrac{4}{5}\right) = \tfrac{12}{5} + \tfrac{20}{5} = \tfrac{32}{5} = 6.4.$$

Solution 5. Steepest ascent points along the gradient itself, $\nabla f(1, 2) = (4, 5)$, i.e. the unit direction $(4, 5)/\sqrt{41}$. The maximum rate of increase is the gradient's magnitude:

$$|\nabla f(1, 2)| = \sqrt{4^2 + 5^2} = \sqrt{41} \approx 6.40.$$

Solution 6. With $z_x = 2x$, $z_y = 2y$, $x' = -\sin t$, $y' = \cos t$:

$$\frac{dz}{dt} = (2x)(-\sin t) + (2y)(\cos t) = 2\cos t(-\sin t) + 2\sin t(\cos t) = 0.$$

Geometrically the path $(\cos t, \sin t)$ rides the unit circle, where $z = x^2 + y^2 = 1$ is constant — so its rate of change is zero, and the velocity is tangent to the level curve, perpendicular to ∇z.

Solution 7. The tree has two branches to s: $\dfrac{\partial z}{\partial s} = z_x\, x_s + z_y\, y_s$ with $z_x = 2xy$, $z_y = x^2$, $x_s = 1$, $y_s = t$:

$$\frac{\partial z}{\partial s} = (2xy)(1) + (x^2)(t) = 2(s + t)(st) + (s + t)^2\, t.$$

Factoring $t(s + t)$ gives

$$\frac{\partial z}{\partial s} = t(s + t)\big[\,2s + (s + t)\,\big] = t(s + t)(3s + t).$$

Solution 8. Set the gradient to zero:

$$f_x = 2x - 4 = 0 \Rightarrow x = 2, \qquad f_y = 2y - 6 = 0 \Rightarrow y = 3,$$

so the only critical point is $(2, 3)$. The second partials are $f_{xx} = 2$, $f_{yy} = 2$, $f_{xy} = 0$, giving

$$D = f_{xx}f_{yy} - (f_{xy})^2 = (2)(2) - 0 = 4 > 0, \quad f_{xx} = 2 > 0,$$

a **local minimum**. The value is $f(2, 3) = 4 + 9 - 8 - 18 + 13 = 0$.

Solution 9. Here $f_x = 2x = 0$ and $f_y = -2y = 0$ give the lone critical point $(0, 0)$. With $f_{xx} = 2$, $f_{yy} = -2$, $f_{xy} = 0$:

$$D = (2)(-2) - 0 = -4 < 0,$$

so $(0, 0)$ is a **saddle point** — f rises along x and falls along y, and the sign of f_{xx} is irrelevant once $D < 0$.

Solution (Capstone). Set both partials to zero:

$$f_x = 3x^2 - 3 = 0 \Rightarrow x = \pm 1, \qquad f_y = 2y = 0 \Rightarrow y = 0,$$

giving critical points $(1, 0)$ and $(-1, 0)$. The second partials are $f_{xx} = 6x$, $f_{yy} = 2$, $f_{xy} = 0$, so $D = 12x$.

$$(1, 0): \ D = 12 > 0, \ f_{xx} = 6 > 0 \Rightarrow \text{local minimum}, \quad f(1, 0) = 1 - 3 = -2.$$

$$(-1, 0): \ D = -12 < 0 \Rightarrow \text{saddle point.}$$

So f has a local minimum value -2 at $(1, 0)$ and a saddle at $(-1, 0)$; the cubic term means there is no global extremum.

Unit 6 Key Takeaway

The workflow is uniform: partials build the gradient ∇f; the gradient drives directional derivatives $D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}$ (always on a *unit* $\mathbf{u}$) and points along steepest ascent $|\nabla f|$; the chain rule sums one term per path; and optimization solves $\nabla f = \mathbf{0}$ then sorts each point with $D = f_{xx} f_{yy} - f_{xy}^2$ — $D > 0$ an extremum (sign of f_{xx} decides min vs. max), $D < 0$ a saddle.